SYED ABDUL RAHMAN ALHADDAD BIN SYED OSMAN ALHADDAD
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PROFILE

Former AI & Data Engineering Intern at Khazanah Nasional Berhad and Computer Science graduate (IIUM) with a proven track record across enterprise data systems, end-to-end ML development, and technical leadership. Backed by a Top 5 national programming finish, 2nd Place FYP commercialization win, and experience co-directing nationwide hackathons, I combine strong engineering execution in automated PySpark pipelines and vector search systems with cross-functional project management. Seeking junior roles in AI Engineering, Data Engineering, and Data Science to deploy data solutions.

CERTIFICATIONS & TECHNICAL SKILLS

- **Certifications: CompTIA Data+** (2025), **Oracle Cloud Infrastructure Generative AI Professional** (2025)
- **Programming:** Python, SQL, TypeScript, JavaScript, R, Java, C++
- **AI & Machine Learning:** RAG, LLMs (Gemini, OpenAI,Anthropic), Sentence-Transformers (GTR-T5), FinBERT, Scikit-learn, TensorFlow, PyTorch, spaCy, HDBSCAN, NLTK, OpenCV, Azure AI Foundry
- **Data Engineering & Analytics:** PySpark, Microsoft Fabric, Pandas, NumPy, Snowflake, BigQuery, Power BI, Tableau, Matplotlib, Seaborn
- **Web, Backend & Scraping:** Next.js, React, Node.js, FastAPI, Tailwind CSS, Prisma ORM, Selenium, BeautifulSoup4, HTML
- **Databases & Cloud:** PostgreSQL (Neon), MongoDB, ChromaDB, FAISS, MySQL, S3 Storage, Oracle Cloud Infrastructure (OCI), Vercel, Microsoft Lake House
- **Testing & Development Tools:** Docker, Playwright, Vitest, Git, GitHub, Linux/WSL, Jupyter, VS Code, C
- **Productivity:** Microsoft Office Suite

PROFESSIONAL EXPERIENCE

Khazanah Nasional Berhad | *AI & Data Engineering Intern*| **March 2026 – September 2026**

- Built resilient API ingestion pipelines in **Python** using back-off retries, state checkpointing, and exception handling to prevent pipeline failures from corrupted symbols and delisted firms.
- Architected a medallion lakehouse pattern (Bronze/Silver/Gold) in Microsoft Fabric with PySpark to clean raw filings and structure analytical datasets for downstream ML workflows
- Developed a corporate sentiment scoring pipeline using **FinBERT** and **spaCy** to turn unstructured reports into quantitative metrics for investment stakeholders.
- Applied **HDBSCAN** to document embeddings generated with **text-embedding-3-large** to identify clusters of emerging financial risks.
- Benchmarked LLMs for automated topic labeling of financial-risk clusters to support investment analysts' document review.
- **Overall Impact:** Modernized investment research by deploying production lakehouse and LLM pipelines that automated report parsing and accelerated due diligence.

PROJECT EXPERIENCE

AI-Powered Personal Shopper Chatbot | *Final Year Project* | **Jan 2026**

- Built a Chrome Extension and real-time stealth scraping pipeline (undetected_chromedriver, BeautifulSoup) to aggregate live Shopee and Lazada product data.
- Engineered a **RAG** backend using FastAPI, MongoDB, and **ChromaDB** with **GTR-T5** embeddings, achieving **88% semantic retrieval accuracy**.

- Integrated **Gemini 2.5 Flash Lite** for intent parsing, context-aware memory, and "Smart Judge" filtering to eliminate irrelevant listings.
- **Result**: Delivered a cross-platform AI shopping assistant with 2.4s backend latency, winning 2nd Place for Best Commercialization.

Full-Stack E-commerce & Fragrance Recommendation Platform | Independent Project | October 2026 – Present

- Built a responsive perfume storefront and admin dashboard using **Next.js, React, TypeScript**, and **Tailwind CSS**.
- Developed an English/Malay scent advisor with deterministic fallback, and weighted ranking constrained by budget, stock availability, and excluded fragrance notes.
- Implemented **PostgreSQL/Prisma** data models, authentication, inventory management, and transactional sandbox checkout.
- Added **Neon** database and S3-compatible image storage with WebP processing.
- Verified local workflows with 17 passing integration tests and 24 passing Playwright browser cases across multiple runs.
- **Result:** Delivered an integrated perfume shopping platform combining personalized fragrance recommendations with centralized product, inventory, and order management.

Predictive & Diagnostic Analysis of Supply Chain Performance | *Academic Project* | **July 2025**

- Analyzed retail and logistics datasets using Pandas, NumPy, Seaborn, and Matplotlib to identify trends in profitability, delivery risk, and regional performance.
- Developed predictive models using **Random Forest, Gradient Boosting, LightGBM, XGBoost, and SVM** to forecast order profit and identify operational inefficiencies.
- **Result:** Produced a diagnostic framework that highlights high-impact factors that supports data-driven improvements in supply chain decision-making.

Hybrid ML Model for Lung Cancer Classification | *Academic Project* | **January 2025**

- Built a hybrid Random Forest and XGBoost ensemble, using Genetic Algorithm and Particle Swarm Optimization for hyperparameter tuning.
- **Result**: Achieved 94% classification accuracy, the highest among the models evaluated in the project.

LEADERSHIP

Deputy Director | I-Hax Innovation Hackathon 2024 | 2024: Co-directed nationwide event for 100+ participants, coordinating operations and securing RM 15k+ in sponsorships.

Head Officer (Entrepreneurship) | ICT Student Society (ICTSS), IIUM | 2022 - 2023: Managed large-scale campus events, generating RM 9,000+ profit and coordinating hardware showcases.

EDUCATION

International Islamic University Malaysia | Oct 2022 – Sept 2026
Bachelor of Computer Science (Data Science & Computational Intelligence) | CGPA: 3.56/4.00

- **Relevant Coursework:** Machine Learning, Natural Language Processing, Data Science, Big Data Analytics, Intelligent Systems.
- **Awards:** Dean's List (5 Semesters) | 2nd Place, FYP Best Commercialization | Top 5, Nationwide Top Coders 2024 | 2nd Place, Top Coders IIUM 2024

Foundation in Engineering & Computer Science | CGPA: 3.37/4.00
Centre for Foundation Studies IIUM | Jul 2021 - Jun 2022

REFERENCES AVAILABLE UPON REQUEST